

ChE 382: Unit Operations Laboratory

EVAPORATION

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Introduction

Evaporation is a process used to concentrate a solution consisting of a nonvolatile solute and a volatile solvent. In most evaporation processes, the solvent is water. In practice, evaporation is conducted by vaporizing a portion of the solvent to produce a concentrated solution or thick liquor. Therefore, this is basically a heat transfer unit operation [1]. In this experiment, a single-effect evaporator is used to evaporate a sugar solution. In order to determine the effect of process parameters such as vacuum pressure, the experiments are carried out at a constant concentration of the solution.

Single-and multiple-effect operation

Most evaporators are heated by steam which condenses on metal tubes. Most often the material to be evaporated flows inside the tubes. Usually the steam is at a low pressure, below 2000 mmHg (40 psia); and often the boiling liquid is under a moderate vacuum, up to about 700 mmHg of vacuum. The vacuum reduces the boiling temperature of the liquid, and increases the temperature difference between the steam and the boiling liquid. This increases the heat-transfer rate in the evaporator.

When a single evaporator is used, the vapor from the boiling liquid is condensed and discarded. This method is called single-effect evaporation, and although it is simple, it utilizes steam ineffectively. About 1 to 1.3 kg of steam is required for each 1 kg of water which evaporates. However, the vapor from the evaporator still contains energy. Therefore, the vapor from one evaporator can be fed into the steam chest of a second evaporator. This vapor can then be sent to a third evaporator, etc. until the vapor has lost a significant amount of energy. This is referred to as multiple-effect evaporation. The heat in the original steam is reused in the subsequent effects, and the evaporation achieved by the original 1 kg of steam is theoretically multiplied by the number of effects. (See *Figure 1*) Notice that the feed to one stage is the product from the previous stage. Also, each effect must operate at a lower pressure to provide for a sufficient driving force for the heat transfer.

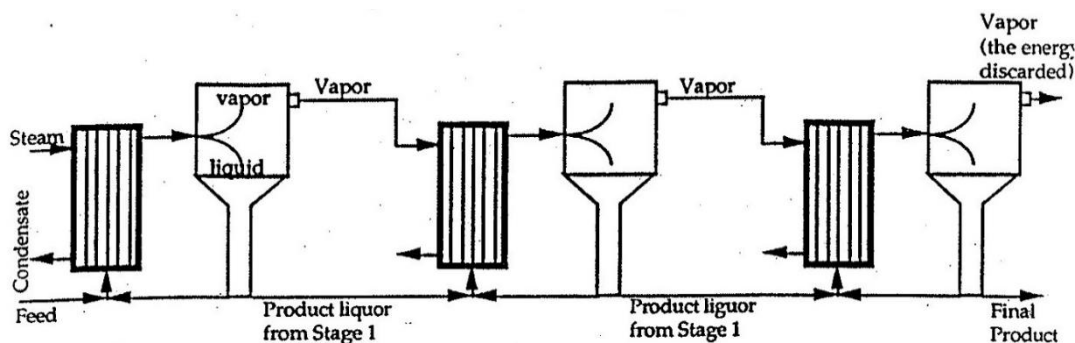


Figure 1: Multiple Effect Evaporation

The principle measures of the performance of a steam-heated tubular evaporator are the **capacity** and the **economy**. Capacity is defined as the number of kilograms of water vaporized of steam fed to the unit and the **economy** as the ratio of (*kg of product/kg of steam used*) In a single-effect evaporator the economy is nearly always less than 1, but in multiple-effect equipment it may be considerably greater. The steam consumption in kg per hour, is also important. It equals the capacity divided by the economy.

Design Equations

The evaporator is basically a heat transfer operation. In order to calculate the capacity and economy, an energy balance must be performed on the evaporator. The following discussion is for a single effect evaporator.

The rate of heat transfer to an evaporator is given by:

$$q = UA\Delta T \quad (1)$$

Where

- U = the overall heat transfer coefficient
- A = the area available for the heat transfer
- ΔT = the average temperature difference for heat transfer (this is normally uniform over the length of the evaporator since one side is condensing steam and the other side is boiling fluid)

If the feed to the evaporator is at the boiling temperature (at the given pressure), all the heat transferred through the heating surface is available for evaporation, and the capacity is proportional to “ q ”. If the feed is cold, the heat required to heat it to its boiling point may be quite large, and the capacity for a given value of “ q ” is reduced accordingly, as heat used to heat the feed is not available for evaporation. Conversely, if the feed is at a temperature above the boiling point in the vapor space, a portion of the feed evaporates spontaneously by adiabatic equilibration with the vapor-space pressure, and the capacity is greater than that corresponding to “ q ”. This process is called flash evaporation. Therefore, “ q ” should be corrected for the feed temperature and converted to the desired units of kg of steam per hour by using the heat of vaporization of the steam at the operating pressure.

The actual temperature drop across the heating surface depends on the solution being evaporated, the difference in pressure between the steam and the boiling liquid, and the depth of liquid over the heating surface. In some evaporators the velocity of the liquid in the tubes also influences the temperature drop, because the frictional loss in the tubes increases the effective pressure of the liquid. When the solution has the characteristics of pure water, its boiling point can be read from the steam tables if the pressure is known, as can the temperature of the condensing steam. In actual evaporators, however the boiling point of a solution is affected by two factors, boiling-point elevation and liquid head.

The economy of an evaporator was defined above. In theory, the economy should equal “1” (unity) if there are no energy losses in the evaporator, and if the feed enters at its boiling point.

Experimental:

A single-effect evaporator is used in this experiment to evaporate a sugar solution. A vacuum pump is connected to the loop containing the solution; this controls the pressure (and hence the boiling point) in the separator. Also, thermocouples are connected at various points. The water vapor is sent to a condenser to collect the water vapor product.

Procedure

- Drain contents of the separator through a valve at the bottom of the separator (it may already be empty).
- Prepare a 10 wt% sugar solution, and fill the separator to the level line. Allow some of the solution to flow into the tube used to drain the separator.
- Put pure deionized water in the makeup tank. IMPORTANT NOTE: During the experiment, you must adjust the flow of the makeup water so that the sugar solution stays at a constant level in the separator.
- Start the flow of steam.
- Start the cooling water to the steam trap. This will condense any remaining steam into large Erlenmeyer Flask.
- Turn on the vacuum pump and adjust the bypass valve to obtain the first desired pressure in the separator (the pressure gauge reads in inches of mercury). After some time, the solution will start circulating through the heat exchanger since a temperature difference now exists for the exchange of heat (Note: this may take some to happen since the solution must heat up to boiling).
- Wait for the system to reach a steady state as indicated by the temperature readings. You will then perform the experiment at this first vacuum pressure.
- After reaching a steady state, empty the steam condensate flask and make sure at least one of the Erlenmeyer flasks for the water product is empty. Begin your measurement of the volume of steam condensate and water product for a given time. Wait for 5-10 minutes and make another measurement at this vacuum pressure. Use the volumes and time to calculate the flowrates of these two streams (i.e., measure the time required to collect a given volume of fluid).
- Record the flow rate of the cooling water to the steam trap, and the inlet and outlet temperature. You need this in order to determine how much of the steam condensed in the heat exchanger (which then provided this energy to the solution).
- Now, change the pump to a different vacuum level, for a total of three vacuum settings. This will change the boiling point of the solution (why) and hence

should increase the temperature difference between the solution and the steam. This should increase the capacity (but what about the economy?).

Data Analysis

Calculate the **capacity** by calculating the amount of product (condensate) collected per time (*kg/hour*) and create a graph of capacity versus absolute pressure.

Calculate the **economy** as the ratio of (*kg of product/kg of steam used*) and create a graph of Economy versus absolute pressure..

Explain any trends you observe in the performance of the evaporator when varying the vacuum setting. For example, what is the temperature difference between the steam and the solution at each vacuum setting? Does the capacity correlate with the temperature difference? How would you use the economy? Also, how does the boiling point of the sugar solution change with vacuum pressure?

Further analysis: Perform an energy balance on the steam and the solution in the evaporator. (I suggest that you draw the appropriate control volumes on some scratch paper in order to do this correctly. Chemical engineers must always know what control volume they are using.)

Further Questions to Consider

1. Pure deionized water is used as the makeup solution to the system. Consequently, the concentration of the solution is not changing during the experiment. Why is this done in these experiments? An industrial evaporator would feed a dilute solution, and continuously remove part of the concentrated solution.
2. Does the heat of vaporization of steam depend on pressure? Where is the heat of vaporization found on a steam table?
3. What other factors besides the vacuum setting would affect the economy or the capacity of the evaporator in this lab? Sugar concentration? Steam flowrate? Liquid level in the separator?

References

1. McCabe W.L., Smith J.C., Harriott P. , "Unit Operations of Chemical Engineering", Chapter 16, 7th Edition, McGraw-Hill, NY, NY, 2005.